Master Thesis Proposal

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Thesis Title

Modeling Psychotic Risk from Cannabis Use: Who Is Most Vulnerable and How?

Research question related to biology, medicine, or health science

How does cannabis use interact with individual genetic risk profiles, adverse childhood experiences (ACE), previous psychotic or manic and depression symptoms, socioeconomic status (SES), sex, and use of other drugs in adolescence to influence the risk of developing psychotic or manic symptoms, and how does it vary across adolescence?

Research question related to biostatistics or data science

Can a Deep Cross-Modal Fusion Network better model complex, non-linear, and holistic interactions between cannabis use, PRS, and other risk factors in predicting adolescence risk of experiencing psychotic or manic symptoms, compared to traditional biostatistical models such as logistic regression or generalized additive model (GAM)?

Background and rationale

Early cannabis use has been linked to a higher risk of developing schizophrenia. Inspired by recent work on adolescent psychedelic use and psychotic symptoms [1], this project focuses on how cannabis use in adolescence relates to psychotic or manic symptoms. Although, initial research shows that people with high polygenic risk scores (PRS) and schizophrenia are more likely to use cannabis, Adorjan et al. [2] suggests that this link could reflect gene–environment correlation.

Motivated by the findings of Yao et al. [3], who showed how schizophrenic genetic risk affects the brain at the level of specific cell types, we plan to compute multiple forms of PRS. These include traditional, pathway-based, and cell-type-specific scores. Since different cell types may matter at different ages, this gives us the opportunity to explore how genetic risk unfolds throughout development in adolescence.

Other risk factors such as Adverse Childhood Experiences (ACE) [4], socioeconomic status (SES), and family history of psychiatric disorders have also been associated with schizophrenic risk [5]. Depressive symptoms are frequently shown to occur during the early course of schizophrenia, appearing as part of a prodromal core syndrome alongside negative symptoms and functional impairment before psychotic symptoms emerge [6]. But most studies look at these factors separately. We still do not fully understand how they combine and interact to influence outcomes.

Psychosis is not caused by any one factor. Instead, it results from the combined effect of many genetic risks and environmental exposures over time [7]. Some risk factors may have more influence at certain ages, for example, childhood trauma might matter more early on, while substance use may be more important later. Understanding these age-specific effects is crucial.

Recently, deep learning methods have shown strong potential in capturing complex relationships. Inspired by this, we propose to apply a Deep Cross-Modal Fusion Network (DCMFNet) [8] to integrate genetic, behavioral, environmental, and demographic data to model schizophrenia risk. This network architecture allows us to explore not only who is most vulnerable, but also how and when various risk factors interact and accumulate during adolescence.

Furthermore, recent benchmarking studies show that deep learning models consistently outperform traditional methods across a wide range of tabular datasets [9], highlighting the evolving strength of these models in handling structured health data. By adopting this flexible and interpretable architecture, our goal is to uncover context-specific patterns in schizophrenia risk that may be missed by linear or rule-based approaches.

Data

This study will use existing data from the Swedish Twin Registry (STR), specifically the Child and Adolescent Twin Study in Sweden (CATSS) cohort at ages 9, 15, and 18. The CATSS datasets consists of approximately 16,000 individuals, including the following wide range of variables relevant to schizophrenia risk:

- **Exposures:** Cannabis use, use of other drugs, and addiction-related behaviors (e.g., smoking, alcohol, snuff) at age 15
- **Environmental factors:** Adverse Childhood Experiences (ACE), including emotional, sexual, and physical abuse; bullying; physical and emotional neglect; witnessing family violence; and hate crimes at age 9 and 15
- **Demographic factors:** Socioeconomic status (SES), including parental education level and country of birth, Sex
- **Previous Symptoms:** Psychotic and manic symptoms, self-reported and parent-reported at age 15 and Depression symptoms, parent-reported at age 15 and self-reported at age 15 and 18
- **Outcomes:** Psychotic and manic symptoms, self-reported and parent-reported at age 18

Genetic Data (PRS):

Polygenic Risk Scores (PRS) for schizophrenia will be computed using genome-wide genotype data from the CATSS cohort, using summary statistics from a large schizophrenia GWAS. These scores will be computed for various forms and included as a genetic matrix input into the model.

Proposed Methods

This project will compare traditional and deep learning models to predict schizophrenia-related symptoms using multiple risk factors.

Baseline Models

A traditional model will be used as a baseline for comparison, either logistic regression or a generalized additive model (GAM), with the final choice guided by initial fit. Logistic regression is widely used in public health due to its simplicity and transparency but assumes linear relationships and limited interaction effects. GAMs, while capable of capturing complex patterns, may lack interpretability and are not specifically tailored for structured multi-modal data. In either case, we will explicitly include interaction terms based on prior literature where appropriate.

Deep Cross-Modal Fusion Network (DCMFNet)

This main model is designed to handle different types of modalities (genetic, behavioral, environmental, demographic) collected at different ages. It uses Iterative Gated Fusion (IGF), which gradually combines information from the different modalities in several layers. In each layer, the model learns which pieces of information are the most important, and refines its understanding of what matters most for predicting symptoms. With each new iteration, the fused features become sharper and more focused, helping the model highlight the most relevant patterns for the final prediction. The design includes:

- Feature encoders for each modality (e.g., PRS, ACE, Cannabis Use)
- A single Iterative Gated Fusion (IGF) block with 5 layers to combine modalities and learn which factors matter more for each individual.
- A final prediction layer that outputs the risk of experiencing schizophrenia symptoms at age 18.

Since much of the data is categorical or comes from questionnaires, using a deep fusion model built for structured, tabular data makes practical and methodological sense.

If training DCMFNet proves computationally infeasible, we will confidently pivot to gradient boosting model CatBoost. It is well-established for structured tabular data and offers strong performance with clear interpretability, making it a reliable primary modeling approach.

Model Comparison

The DCMFNet will be compared to a baseline model on predictive performance (accuracy, F1 score, AUC), ability to model interactions and non-linear effects using SHAP analysis, and interpretability in terms of identifying which features matter most. This comparison will help assess how well each model handles complex data and provides actionable insights for public health research.

PRS Pipeline

PRS will be calculated using genotype data from the CATSS study and schizophrenia GWAS summary statistics. Following the approach of [3], we will compute different types of PRS (e.g., cell-type-specific, such as subregions in the hippocampus and amygdala) and use them as genetic input to DCMFNet. The PRS pipeline includes quality control (PLINK) and score construction.

Work and time plans

The thesis will follow a structured timeline, as shown in Figure 1.

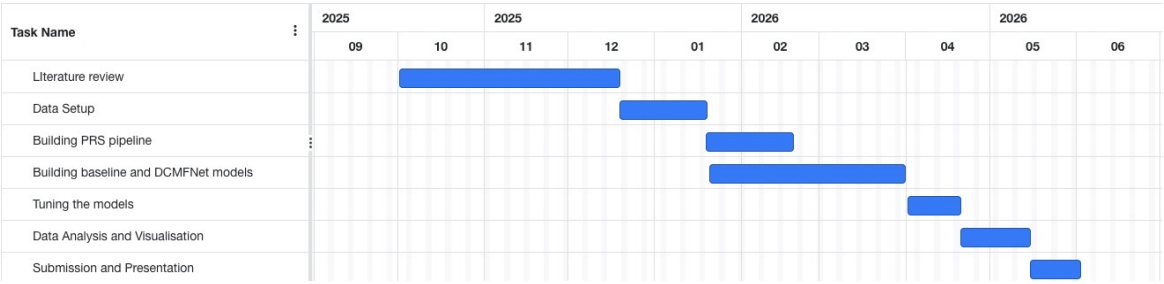

Figure 1: Thesis Timeline

The project started with reading research papers to find important risk factors and the right modeling approach. I have made a list of variables from the Swedish Twin Registry and will soon request genotype data to build the polygenic risk scores (PRS). Next, I will prepare the dataset and build a baseline model (either logistic regression or GAM) along with the DCMFNet before the mid-term seminar. After that, I will focus on improving the models, testing them, analyzing the results, and writing the thesis report. I will meet weekly with my supervisors for regular guidance.

Ethical Permits

The CATSS-study was approved by the Swedish ethical review authority (Dnr 02-289, 03-672, 2010/597-31/1, and 2010/322-31/2, 2010/1114-32, 2011/828-32, 2016/2135-31).

References

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